

EXPLAINING DEFEASIBLE REASONING

A step-through pedagogical tool for defeasible entailment

Try the web app!

1 Background

Classical reasoning treats rules as having no exceptions. This becomes a problem when general rules conflict with special cases.

Defeasible reasoning allows rules such as “platypuses typically do not wear hats” while still allowing Perry to be an exception.

2 Objectives

Develop an interactive tool that shows, step by step, how different defeasible reasoning algorithms **handle exceptions** and reach a conclusion - making it easier to see where they differ and why.



3 How it works

- Enter rules & a question
- Choose a reasoning algorithm (RC, LC, RelC)
- Ranking the information
- Follow the reasoning algorithm step by step
- Compare the results

4 Meet Perry

Consider a knowledge base about Perry the platypus. Perry wears a hat, making him an exception. We rank our rules based on how specific they are. More specific rules are favoured when there is an exception. Using our defeasible algorithms, what can we conclude about Perry?

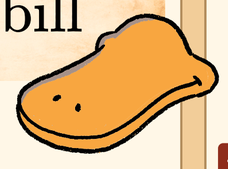
Knowledge base

$\text{perry} \rightarrow \text{platypus}$	(Perry is a platypus)
$\text{platypus} \mid\sim \neg\text{hat}$	(Platypuses typically do not wear hats)
$\text{platypus} \mid\sim \text{bill}$	(Platypuses typically have bills)
$\text{perry} \mid\sim \text{hat}$	(Perry typically wears a hat)

Query

Does Perry typically have a bill?

$\text{perry} \mid\sim \text{bill}$



5 Results

PDR decomposes defeasible algorithms into their intermediate reasoning steps, explaining why the same query gets different answers.

Rational Closure

Exposes how whole ranks of rules are removed when a conflict occurs.

NO

Lexicographic Closure

Exposes which subsets of the ranks are tested and why the algorithm keeps reducing the information considered

YES

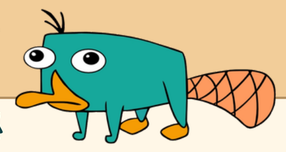
Relevant Closure

Exposes how relevant information is identified and used to resolve the conflict.

YES

- ✓ The answers matched theoretical examples and existing implementations.
- ✓ The steps shown by the tool accurately explain the reasoning process.
- ✓ The intermediate states are made visible

6 What is PDR?



PDR is an interactive web-based pedagogical tool for exploring defeasible reasoning. Rather than showing only a final answer, it exposes the intermediate steps used to reach an answer.

Step Through

Follow the reasoning process step-by-step, with rankings, pseudocode, and plain-English explanations.

Compare

See how the same knowledge base and query are handled differently by Rational Closure, Lexicographic Closure, and Relevant Closure.

Understand

Explore reasoning through visualisations, understandable explanations, and supporting resources.

7 Conclusions



PDR visualises and explains the defeasible reasoning process. PDR exposes intermediate reasoning states into clear, interactive steps, the tool allows users to see how the different approaches handle the same query and why they reach different conclusions.

Project Team:
Samukelisiwe Zwane | ZWNSAM011@myuct.ac.za
Liam De Saldanha | DSLLIA002@myuct.ac.za
Nikita Martin | MRTNIK003@myuct.ac.za

Supervised by:
Prof. Tommie Meyer | tmeyer@cair.org.za

Department of Computer Science
Tel: 27(0)21 650 2663
Website: https://sit.uct.ac.za/

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